

A Detailed Mathematical Guide to Principal Component Analysis (PCA)

Contents

1	The Intuition: Why Do We Need PCA?	2
2	Developing the Mathematical Intuition: From Variance to Eigenvectors	2
2.1	1. The Goal: Maximizing Variance	2
2.2	2. How to Measure Variance in a Specific Direction?	2
2.3	3. The Emergence of the Covariance Matrix	3
2.4	4. Connecting to Eigendecomposition	3
3	The Two Sides of PCA: The Core Optimization Problems	4
3.1	Perspective 1: Maximizing Variance	4
3.2	Perspective 2: Minimizing Reconstruction Error	4
4	Detailed 3D Example: From Data to Projection	5
4.1	Original Data Matrix (X)	5
4.2	Step 1: Mean Center the Data	5
4.3	Step 2: Calculate the 3x3 Covariance Matrix	5
4.4	Step 3: Find Eigenvalues and Eigenvectors	5
4.5	Step 4: Project the Data onto the Principal Components	6

1 The Intuition: Why Do We Need PCA?

Imagine you have a dataset with many features—say, hundreds or even thousands. Trying to visualize or model such high-dimensional data is incredibly difficult. Often, many of these features are redundant or correlated. For example, in a dataset about houses, features like "square footage" and "number of bedrooms" are likely to be highly correlated.

Principal Component Analysis (PCA) gives us a way to tackle this complexity. Its core idea is to change the coordinate system in which we view our data. Instead of using the original feature axes, PCA finds a new set of axes, called **principal components**, that are better aligned with the "shape" of the data cloud.

What makes these new axes "better"?

1. **They capture the maximum variance:** The first principal component (PC1) is a line drawn through the data that captures the most possible variation. The second principal component (PC2) is perpendicular to the first and captures the next most variation, and so on.
2. **They are uncorrelated:** Each principal component is orthogonal (at a right angle) to the others, meaning the new "features" are not redundant.

By using only the first few principal components, we can often represent the data in a much lower-dimensional space while losing very little of the essential information (i.e., the variance). This makes PCA a powerful tool for dimensionality reduction, data visualization, and noise filtering.

2 Developing the Mathematical Intuition: From Variance to Eigenvectors

The entire method of PCA is born from a single, clear goal: find the direction in the data with the most "spread" or variance. Let's walk through how this goal mathematically leads us directly to the concepts of covariance matrices and eigenvectors.

2.1 1. The Goal: Maximizing Variance

Let's formalize our setup. Our data is a matrix X of size $n \times d$, where n is the number of data points and d is the number of features. The first step, which is crucial, is to mean-center the data by subtracting the mean of each feature. We'll call this centered data matrix $\hat{X}$.

We are looking for a direction in our d -dimensional space. Any direction can be represented by a unit vector u (with dimension $d \times 1$). We use a unit vector because we only care about the direction, not its magnitude, so we enforce the constraint that $u^T u = 1$.

2.2 2. How to Measure Variance in a Specific Direction?

To find the variance of our data along the direction u , we first need to project each data point (each row of $\hat{X}$) onto this direction. The projection of a single data point $\hat{x}_i$ (a row vector) onto the direction vector u is given by the dot product: $\hat{x}_i u$.

If we do this for all n data points, we get a new vector of projected values: $P = \hat{X}u$. This vector P is of size $n \times 1$, and it represents our entire dataset transformed into a single dimension along the direction u .

Now, we calculate the variance of this new 1D dataset, P . Since the data was mean-centered, the variance is simply the average of the squared values (with a factor of $n - 1$ for the sample

variance):

$$\text{Var}(P) = \frac{1}{n-1} \sum_{i=1}^n P_i^2$$

In matrix notation, this sum of squares is simply $P^T P$. So:

$$\text{Var}(P) = \frac{1}{n-1} P^T P$$

2.3 3. The Emergence of the Covariance Matrix

This is the key step. Let's substitute our expression for P back into the variance formula:

$$\text{Var}(\hat{X}u) = \frac{1}{n-1} (\hat{X}u)^T (\hat{X}u)$$

Using the rule of transposes $(AB)^T = B^T A^T$, we get:

$$\text{Var}(\hat{X}u) = \frac{1}{n-1} (u^T \hat{X}^T) (\hat{X}u)$$

Rearranging the brackets (since matrix multiplication is associative):

$$\text{Var}(\hat{X}u) = u^T \left(\frac{1}{n-1} \hat{X}^T \hat{X} \right) u$$

Look closely at the term in the parentheses. This is precisely the mathematical definition of the sample **covariance matrix** of our data!

$$C = \frac{1}{n-1} \hat{X}^T \hat{X}$$

So, the variance of the data projected onto any direction u is given by the expression:

$$\text{Var}(\text{projected data}) = u^T C u$$

2.4 4. Connecting to Eigendecomposition

Our problem has now been transformed into a formal optimization problem:

$$\underset{u}{\text{maximize}} \quad u^T C u \quad \text{subject to} \quad u^T u = 1$$

This is a classic constrained optimization problem that can be solved using Lagrange multipliers. The solution reveals that the vectors u that maximize this expression are the **eigenvectors** of the matrix C .

Let's see why. The equation for an eigenvector v and eigenvalue λ of a matrix C is:

$$Cv = \lambda v$$

Now, let's pre-multiply both sides by v^T :

$$v^T C v = v^T (\lambda v) = \lambda (v^T v)$$

Since eigenvectors are chosen to be unit vectors, we have $v^T v = 1$. This leaves us with:

$$v^T C v = \lambda$$

This is the punchline! This equation tells us that the variance of the data projected onto an eigenvector v is simply its corresponding eigenvalue λ . To find the direction of maximum variance, we just need to find the eigenvector of the covariance matrix that has the largest eigenvalue. This is the first principal component.

3 The Two Sides of PCA: The Core Optimization Problems

As we've just seen, the mathematical foundation of PCA can be derived directly from the goal of maximizing variance. This leads to one of two equivalent ways to frame the core optimization problem.

3.1 Perspective 1: Maximizing Variance

- **The Optimization Problem:**

$$\underset{u}{\text{maximize}} \quad u^T C u \quad \text{subject to} \quad u^T u = 1$$

The solution is that the optimal vectors u are the eigenvectors of the covariance matrix C .

3.2 Perspective 2: Minimizing Reconstruction Error

This perspective asks how to project data to a lower dimension while minimizing the information loss, measured by reconstruction error. To see how this connects to maximizing variance, let's derive it.

The error for a single data point $\hat{x}_i$ (treated as a column vector) is the squared distance between the original point and its reconstruction:

$$\text{Error}_i = \|\hat{x}_i - (\hat{x}_i^T u)u\|^2$$

Let's expand the squared norm, where $(\hat{x}_i^T u)$ is a scalar projection:

$$\begin{aligned} \text{Error}_i &= (\hat{x}_i - (\hat{x}_i^T u)u)^T (\hat{x}_i - (\hat{x}_i^T u)u) \\ &= (\hat{x}_i^T - (\hat{x}_i^T u)u^T)(\hat{x}_i - (\hat{x}_i^T u)u) \\ &= \hat{x}_i^T \hat{x}_i - \hat{x}_i^T (\hat{x}_i^T u)u - (\hat{x}_i^T u)u^T \hat{x}_i + ((\hat{x}_i^T u)u^T)((\hat{x}_i^T u)u) \\ &= \hat{x}_i^T \hat{x}_i - (\hat{x}_i^T u)^2 - (\hat{x}_i^T u)^2 + (\hat{x}_i^T u)^2(u^T u) \end{aligned}$$

Since u is a unit vector, $u^T u = 1$. The expression simplifies to:

$$\text{Error}_i = \hat{x}_i^T \hat{x}_i - (\hat{x}_i^T u)^2$$

Our goal is to minimize the total error over all points, $J = \sum_{i=1}^n \text{Error}_i$:

$$J = \sum_{i=1}^n (\hat{x}_i^T \hat{x}_i - (\hat{x}_i^T u)^2) = \sum_{i=1}^n \hat{x}_i^T \hat{x}_i - \sum_{i=1}^n (\hat{x}_i^T u)^2$$

The first term, $\sum \hat{x}_i^T \hat{x}_i$, is the total variance of the data, which is a constant value with respect to our choice of u . Therefore, to **minimize** the total error J , we must **maximize** the second term, $\sum (\hat{x}_i^T u)^2$.

This second term is the sum of the squared projected lengths. This is exactly proportional to the variance of the projected data we sought to maximize in the first perspective!

$$\sum_{i=1}^n (\hat{x}_i^T u)^2 = \sum_{i=1}^n (u^T \hat{x}_i)(\hat{x}_i^T u) = u^T \left(\sum_{i=1}^n \hat{x}_i \hat{x}_i^T \right) u = u^T (\hat{X}^T \hat{X}) u = (n-1)u^T C u$$

Thus, minimizing reconstruction error is equivalent to maximizing variance.

- **The Optimization Problem:**

$$\underset{u}{\text{minimize}} \quad \sum_{i=1}^n \|\hat{x}_i - (\hat{x}_i^T u)u\|^2 \quad \text{subject to} \quad u^T u = 1$$

Both paths lead to the same destination: the solution is to choose the eigenvectors of the covariance matrix as the principal components.

4 Detailed 3D Example: From Data to Projection

Let's walk through a concrete example with 3 features ($d = 3$) and 5 data points ($n = 5$).

4.1 Original Data Matrix (X)

Suppose our data measures three features (e.g., Height, Weight, Age):

$$X = \begin{pmatrix} 2 & 4 & 5 \\ 3 & 6 & 7 \\ 4 & 5 & 6 \\ 5 & 7 & 8 \\ 6 & 8 & 8 \end{pmatrix}$$

4.2 Step 1: Mean Center the Data

First, calculate the mean of each column:

- $\mu_1 = (2 + 3 + 4 + 5 + 6)/5 = 4.0$
- $\mu_2 = (4 + 6 + 5 + 7 + 8)/5 = 6.0$
- $\mu_3 = (5 + 7 + 6 + 8 + 8)/5 = 6.8$

Now, subtract the means to get the centered matrix $\hat{X}$:

$$\hat{X} = \begin{pmatrix} -2 & -2 & -1.8 \\ -1 & 0 & 0.2 \\ 0 & -1 & -0.8 \\ 1 & 1 & 1.2 \\ 2 & 2 & 1.2 \end{pmatrix}$$

4.3 Step 2: Calculate the 3x3 Covariance Matrix

The covariance matrix is $C = \frac{1}{n-1} \hat{X}^T \hat{X}$. Here, $n - 1 = 4$.

$$C = \frac{1}{4} \begin{pmatrix} -2 & -1 & 0 & 1 & 2 \\ -2 & 0 & -1 & 1 & 2 \\ -1.8 & 0.2 & -0.8 & 1.2 & 1.2 \end{pmatrix} \begin{pmatrix} -2 & -2 & -1.8 \\ -1 & 0 & 0.2 \\ 0 & -1 & -0.8 \\ 1 & 1 & 1.2 \\ 2 & 2 & 1.2 \end{pmatrix}$$

The resulting covariance matrix is:

$$C = \frac{1}{4} \begin{pmatrix} 10 & 9 & 7 \\ 9 & 10 & 8 \\ 7 & 8 & 6.8 \end{pmatrix} = \begin{pmatrix} 2.5 & 2.25 & 1.75 \\ 2.25 & 2.5 & 2.0 \\ 1.75 & 2.0 & 1.7 \end{pmatrix}$$

4.4 Step 3: Find Eigenvalues and Eigenvectors

To find the eigenvalues, we must solve the characteristic equation $\det(C - \lambda I) = 0$.

$$\det \begin{pmatrix} 2.5 - \lambda & 2.25 & 1.75 \\ 2.25 & 2.5 - \lambda & 2.0 \\ 1.75 & 2.0 & 1.7 - \lambda \end{pmatrix} = 0$$

Expanding this determinant yields a cubic polynomial in λ :

$$(2.5 - \lambda)[(2.5 - \lambda)(1.7 - \lambda) - 4] - 2.25[2.25(1.7 - \lambda) - 3.5] + 1.75[4.5 - 1.75(2.5 - \lambda)] = 0$$

Solving this polynomial analytically is highly complex. For practical purposes, numerical methods are used. These methods give us the eigenvalues:

- $\lambda_1 \approx 6.43$
- $\lambda_2 \approx 0.22$
- $\lambda_3 \approx 0.05$

To find the eigenvector v_1 corresponding to $\lambda_1 \approx 6.43$, we solve $(C - \lambda_1 I)v_1 = 0$:

$$\begin{pmatrix} 2.5 - 6.43 & 2.25 & 1.75 \\ 2.25 & 2.5 - 6.43 & 2.0 \\ 1.75 & 2.0 & 1.7 - 6.43 \end{pmatrix} \begin{pmatrix} v_{11} \\ v_{12} \\ v_{13} \end{pmatrix} = \begin{pmatrix} -3.93 & 2.25 & 1.75 \\ 2.25 & -3.93 & 2.0 \\ 1.75 & 2.0 & -4.73 \end{pmatrix} \begin{pmatrix} v_{11} \\ v_{12} \\ v_{13} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

This gives us a system of linear equations. Solving it (e.g., using Gaussian elimination) reveals the relationships between the components, such as $v_{12} \approx 1.05v_{11}$ and $v_{13} \approx 0.88v_{11}$. We can choose $v_{11} = 1$ to get a vector $(1, 1.05, 0.88)$. To make it a unit vector, we divide by its magnitude $\sqrt{1^2 + 1.05^2 + 0.88^2} \approx 1.69$.

$$v_1 \approx \frac{1}{1.69} \begin{pmatrix} 1 \\ 1.05 \\ 0.88 \end{pmatrix} = \begin{pmatrix} 0.59 \\ 0.62 \\ 0.52 \end{pmatrix}$$

Repeating this process for λ_2 and λ_3 gives the other eigenvectors:

- $v_2 \approx \begin{pmatrix} -0.79 \\ 0.21 \\ 0.58 \end{pmatrix}$
- $v_3 \approx \begin{pmatrix} 0.18 \\ -0.76 \\ 0.63 \end{pmatrix}$

4.5 Step 4: Project the Data onto the Principal Components

Let's reduce our data from 3D to 2D by projecting it onto the first two principal components. We form a projection matrix V whose columns are the first two eigenvectors:

$$V = (v_1 \ v_2) = \begin{pmatrix} 0.59 & -0.79 \\ 0.62 & 0.21 \\ 0.52 & 0.58 \end{pmatrix}$$

The final projected data, Z , is obtained by multiplying our centered data $\hat{X}$ by this projection matrix V :

$$Z = \hat{X}V = \begin{pmatrix} -2 & -2 & -1.8 \\ -1 & 0 & 0.2 \\ 0 & -1 & -0.8 \\ 1 & 1 & 1.2 \\ 2 & 2 & 1.2 \end{pmatrix} \begin{pmatrix} 0.59 & -0.79 \\ 0.62 & 0.21 \\ 0.52 & 0.58 \end{pmatrix} = \begin{pmatrix} -3.36 & 0.12 \\ -0.49 & 0.91 \\ -1.04 & -0.67 \\ 1.83 & 0.10 \\ 2.98 & -0.46 \end{pmatrix}$$

This 5×2 matrix Z is our final result. We have successfully transformed our original 3-dimensional data into a new 2-dimensional representation that captures the maximum possible variance (approximately $\frac{\lambda_1 + \lambda_2}{\lambda_1 + \lambda_2 + \lambda_3} \approx 99.2\%$ of the total variance).